Week 10: 2/11 – 6/11

Subject: Math - Statistics **Unit Title: Unit 2: Describing and Displaying Data** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Chebyshev's Theorem	• I can use Chebyshev's theorem to determine the minimum proportion of data within a specified number of standard deviations of the mean.	1.2 Alternatives for auditory information (captions, transcripts, visuals). 8.3 Foster collaboration and community with structured group roles and supports. Choose an item.	Links to support learning: Chebyshev's Theorem - YouTube Standard Deviation Formula, Statistics, Variance, Sample and Population Mean - YouTube
Lesson 2: The Empirical (Normal) Rule	• I can apply the empirical rule to estimate percentages of data within specified intervals of an approximately normal distribution.		
Lesson 3: Standardizing Data Values with z Scores	• I can calculate and interpret z scores to describe and compare the relative positions of values within their datasets.		
Lesson 4: Percentiles and Percentile Graphs	• I can calculate percentile ranks and interpret percentile notation to describe a value's relative position within a dataset.		
Homework	• Complete the tasks in your Notebook.	8.1 Make goals clear, visible, and supported by short-term objectives. Choose an item.	
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments.		

Technology Integration	Microsoft (Power point, excel, etc.)

Cross-curricular links	**English / Media Literacy:** Students interpret survey questions, headlines, and data-based claims, then communicate and justify conclusions using clear evidence.
External Assessment Integration	SAT
Wellbeing/ ATL Skills	**Thinking Skills – Critical Thinking:** Students analyse data sources, identify bias, classify information, and evaluate whether conclusions are supported by the data. **Communication Skills:** Students discuss findings, justify choices, and use appropriate statistical vocabulary. **Research Skills:** Students plan data collection and evaluate the reliability and relevance of information.
Standards	• **HSS-ID.A.4 -** Use the mean and standard deviation of an approximately normal distribution to estimate population percentages and recognize when a normal model is inappropriate. Directly addressed through the empirical rule; z scores provide supporting skills.
Inquiry/ TOK Question	**How can we collect and evaluate data to make reliable conclusions about a population?**
Success Criteria (Measurable)	**Lesson 1: Chebyshev's Theorem** • **Must:** I must be able to calculate the minimum percentage of data within a specified number of standard deviations using Chebyshev's theorem. • **Should:** I should be able to calculate and interpret the corresponding interval around the mean. • **Could:** I could evaluate a claim by distinguishing the theorem's minimum guaranteed percentage from the actual percentage. **Lesson 2: The Empirical Rule** • **Must:** I must be able to identify the intervals containing approximately sixty-eight percent, ninety-five percent, and ninety-nine-point seven percent of the data in an approximately normal distribution. • **Should:** I should be able to calculate interval endpoints within one, two, and three standard deviations of the mean. • Could: I could use symmetry to determine percentages between selected interval boundaries or in the tails, explaining why the rule is appropriate. **Lesson 3: Standardizing Data Values with z Scores** • **Must:** I must be able to calculate a z score by subtracting the mean from a data value and dividing by the standard deviation. • **Should:** I should be able to interpret a z score as the number of standard deviations a value lies above or below the mean.

	• **Could:** I could compare values from different distributions using z scores and justify which is more unusual relative to its own dataset. **Lesson 4: Percentiles and Percentile Graphs** • **Must:** I must be able to interpret a percentile rank in context. • **Should:** I should be able to calculate a value's percentile rank using the number of data values below it. • **Could:** I could evaluate a statement about percentile rank and explain the difference between a percentile and a percentage score.
National Identity Domain	Volunteering التطوع Global Understanding التفاهم العالمي Heritage التراث
	Area: Community Responsibility and Informed Decision-Making
	Implementation: Using an anonymized or simulated dataset of daily household water consumption in a UAE community, students apply Chebyshev's theorem to establish a minimum percentage within an interval, use the empirical rule if the distribution is approximately normal, calculate a household's z score, and determine its percentile rank. Students use their findings to propose a water-conservation message that promotes responsibility toward the UAE's natural resources.